

# Roshan Kern

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## EDUCATION

**Massachusetts Institute of Technology** | Cambridge, MA

Aug 2026 ~ 2030

*Ph.D. in Biological Engineering (Computational Focus)*

Accepted for 2025 cohort, deferred for personal reasons

**Case Western Reserve University**, College of Arts and Sciences | Cleveland, OH

Sep 2021 - Dec 2024

*B.S. in Systems Biology (Bioinformatics Subspecialty)*

Minors in Applied Data Science, Computer Science, Business Management

GPA 4.0

## EXPERIENCE

**Telora Fellowship**

Jan 2026 - July 2026

Telora Fellow

*Mentor: Eliam Medina*

- \$50k angel investment and mentorship to start a startup
- Identified problems by interviewing ~100 people per week to establish clear value loss in time, money, or accuracy
- Fractional COO for legal tech startup Hourglass ([hourglass.law](https://hourglass.law)) with \$250k ARR and 18% MoM growth

**Massachusetts Institute of Technology** | Cambridge, MA

Sep 2024 - June 2025 (Jan-June 2025 On-Site)

Research Intern

*Advisor: Iain Cheeseman*

- Co-founded computational team to make the lab cross-functional, implementing policies for project management, code review, and computational workflows.
- Developed Briefflow for processing optical pooled screen data at scale with a focus on modularity/extensibility for effective process development.
- Reduced screening time by 80% (5→1 day) and module integration time by 93% (2 weeks→1 day)

**Fibrolytix Bio** | Denver, CO

Jun 2025 - Aug 2025

CTO

- Co-founded drug discovery company aiming to use proprietary SOTA assay in lab-in-the-loop agentic system
- Developed agent for predicting efficacy of compound for reversing cardiac fibrosis with DSPy
- Reduced agent run costs by 50%

**Recursion Pharmaceuticals** | Salt Lake City, UT

May 2024 - Aug 2024

Data Science Intern

*Mentor: Kyle Hansen*

- Established metric for and optimized gene target selection from large-scale gene-chemical interactions.
- Investigated contribution of embeddings from three phenomic maps to compound selection process.

**Case Western Reserve University** | Cleveland, OH

May 2023 - Aug 2023

Summer Undergraduate Research Program Fellow

*Advisor: Thomas LaFramboise*

- Implemented pipeline for human-endogenous retrovirus profiling from patient transcriptomics data.
- Investigated effect of human-endogenous retrovirus expression on bone marrow failure syndromes.

**University of Colorado Anschutz** | Aurora, CO

May 2022 - May 2023

Research Intern

*Advisor: Gregory Way*

- Trained generalizable machine learning model for classification of single-cell images into one of 16 stages with phenotype features.
- Achieved F1 score of 0.877 on test set and applied model to different cell image datasets to understand generalizability.
- Developed cell image processing tool for extracting features from single-cell images (see projects).

## PUBLICATIONS

4. Nguyen, A.D.D., Kern, R., Dombrovski, T., LaFramboise, T. (2026) The circulating metatranscriptome in acute myeloid leukemia patients. *Cell Reports (in review)*.

3. Di Bernardo, M., **Kern, R.**, Mallar, A., Nutter-Upham, A., Blainey, P.C., Cheeseman, I. (2025) BrieFlow: An integrated computational pipeline for high-throughput analysis of optical pooled screening data. *Nature Communications*: [doi.org/10.1101/2025.05.26.656231](https://doi.org/10.1101/2025.05.26.656231)

2. Serrano, E., Chandrasekaran, S.N., Buntin, D., Brewer, K.I., Tomkinson, J., **Kern, R.**, Bornholdt, M., Fleming, S., Pei, R., Arevalo, J., Tsang, H., Rubinetti, V., Tromans-Coia, C., Becker, T., Weisbart, E., Bunne, C., Kalinin, A.A., Senft, R., Taylor, S.J., Jamali, N., Adeboye, A., Abbasi, H.S., Goodman, A., Caicedo, J.C., Carpenter, A.E., Cimini, B.A., Singh, S., Way, G.P. (2024) Reproducible image-based profiling with Pycytominer. *Nat Methods*: [doi.org/10.1038/s41592-025-02611-8](https://doi.org/10.1038/s41592-025-02611-8)

1. Tomkinson, J.\*, **Kern, R.\***, Mattson, C., Way, G.P. (2024) Toward generalizable single-cell phenotype prediction from nucleus morphology representations. *BMC Methods*: [doi.org/10.1186/s44330-024-00014-3](https://doi.org/10.1186/s44330-024-00014-3), *\*Co-first authors*

## MAJOR PRODUCTS

**Compound Prioritization Agent:** Prioritize compounds for screening in SOTA cardiac fibrosis-reversal screen using previous assay results. ([github.com/FibrolytixBio/cf-compound-selection-demo](https://github.com/FibrolytixBio/cf-compound-selection-demo))

**BrieFlow:** Open source, extensible tool for processing data from Optical Pooled Screens. ([github.com/cheeseman-lab/brieflow](https://github.com/cheeseman-lab/brieflow), [anthropic.com/news/accelerating-scientific-research](https://anthropic.com/news/accelerating-scientific-research)) <https://www.anthropic.com/news/accelerating-scientific-research#cheeseman-lab-automating-the-interpretation-of-large-scale-gene-knockout-experiments>

**PyCytominer:** Open source suite of common functions used to process high dimensional readouts from high-throughput cell experiments. ([github.com/cytomining/pycytominer](https://github.com/cytomining/pycytominer))

**IDR Stream:** Open source software to handle cell image downloading, preprocessing, cell segmentation, feature extraction, feature compilation for Image Data Resource datasets. ([github.com/WayScience/IDR\\_stream](https://github.com/WayScience/IDR_stream))

## HONORS AND FELLOWSHIPS

|      |                                                                                      |
|------|--------------------------------------------------------------------------------------|
| 2026 | MIT Presidential Fellowship                                                          |
| 2024 | Case Alumni Association Junior Senior Scholar                                        |
| 2024 | Dean's High Honors List, Case Western Reserve University                             |
| 2023 | Cancer-focused Summer Undergraduate Research Fellow, Case Western Reserve University |

## PRESENTATIONS AND TEACHING

**CWRU CanSUR Symposium:** “Human Endogenous Retrovirus Landscape of Aplastic Anemia”, *August 2023*

**Way Lab:** Led lab discussions of self-supervised machine learning for live cell imagery segmentation (Robitaille et al., 2022) and the development of a dynamic protein atlas of human cell division (Cai et al., 2018).

**Recursion Pharmaceuticals:** Presented work on gene and compound selection processes to senior data scientists at data analysis reviews.

## COMMUNITY

**Campus Flavor** | Cleveland, OH

Jan 2023 - May 2023

*Project Leader/Coordinator*

- Led two technical developers to create a website for collecting university-specific merchandise designs.
- Received 32 student-created designs in two weeks of hosting competition on the website.
- Communicated with business representatives from API company and university to formulate business strategy.

**Saturday Tutoring Program** | Cleveland, OH

Jan 2023 - May 2023

*Middle School Tutor*

- Tutored middle school students in STEM topics including biology, chemistry, and physics.

**Animal Protective League** | Cleveland, OH

Sep 2023 - Dec 2023

*Dog Walker*

- Walked and played with dogs at APL animal shelter.

**MedWish** | Cleveland, OH

Jan 2022 - May 2023

*Member*

- Organized and inventoried donated medical supplies for distribution to areas in need.
- Led eight novice programmers in building a club website to attract new members.